

Kurtosis — degree of peakedness.

$$\beta_2 = \frac{m_4}{m_2^2} \quad \text{Pearson's beta coefficient}$$

$$\beta_2 = 3$$

$$\beta_2 > 3$$

$$\beta_2 < 3$$

normal curve

+ve Kurtosis (lepto-kurtic)

-ve Kurtosis (platy-kurtic)

$$\gamma_2 = \beta_2 - 3$$

$$\begin{aligned} \gamma_2 = 0 & \quad \text{normal curve} \\ \gamma_2 > 0 & \quad \text{+ve Kurtosis} \\ \gamma_2 < 0 & \quad \text{-ve "} \end{aligned}$$

Numerical problems

①

○ group 1	$n_1, \bar{x}_1$	○ group = group 1 + group 2 $n_1 + n_2 = n$
○ group 2	$n_2, \bar{x}_2$	

$$\bar{x} = ? \quad \bar{x} = \frac{(n_1 \bar{x}_1 + n_2 \bar{x}_2)}{(n_1 + n_2)}$$

②. Sd of 32 obs is 5. The sum of obs is 80, what is the sum of squares of these obs?

— sol: $\sigma^2 = \text{Var}(x), \sigma = \text{Sd}$

$$\sigma^2 = 5^2$$

$$\sigma^2 = \frac{1}{n} \sum x_i^2 - (\bar{x})^2 = 25$$

$$\sum x_i^2 = 1000.$$

∴ All mean and Sd of 20 items are found to

Ex 2. The mean and sd of $\underline{x}$ be 10 and $\underline{2}$ respectively.
 We notice that one item whose value is 8 was incorrect and the actual value is 12.
 What is the corrected mean and sd?

— new mean = 10.2
 new var = 3.96

Ex. 3 In a group of 1000 people, the monthly income of 4% are below 60k,
 15% have income below 62.5k,
 15% earn 95k or above,
 5% earn 100k or above.

Find out the median income.

(ans - 78.75 k)